

Genetic report · longevity

Cancer Risk (Polygenic Scores)

Colorectal, prostate, skin, pancreas, lung and other cancers · implications for screening

Sérgio F · Report date: 02-10-2026

Local DNA analysis (chip + high-confidence genomic imputation) with calibrated polygenic scores in the reference population. Conclusions based only on genes, divided into favourable and unfavourable and linked to 84 aging mechanisms. It is neither a diagnosis nor a prescription. All actions are marked as suggestion (not applied).

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Summary

57

Curated SNPs

54

in the chip

1

imputed
($DR2 \geq 0.8$)

2

without chip
genotype/
imputation

22

Polygenic scores

This report combines polygenic 22 scores (PGS) of cancer, calculated with the DNA of the chip and high confidence imputation and compared with the reference population, with cured 57 SNPs (high penetrance variants and the main GWAS loci). In the end there is a table of implications for screening.

In 4 lines: (1) There is no confirmed pathogenic variant in the classical hereditary cancer genes that the chip covers (BRCA, Lynch, APC, MUTYH, CHEK2, HOXB13, TP53, CDKN2A); (2) the polygenic background is low for melanoma, squamous cell carcinoma, kidney, lymphomas, lung and prostate (complete genomic score p0.9); (3) is high for colorectal (complete genomic score p94.3 and p78.5; early onset p82.3; polyps p92.7), bladder (percentile 90, with slow NAT2), glioma (91, alleles TERT/RTEL1), pancreas (83) and esophagus (85); (4) alleles of the TERT locus that extend telomeres are the same ones that increase the risk of several cancers and clonal hematopoiesis: reinforce caution with telomerase activators.

The percentile is a relative position: an the 90th percentile in a rare cancer (glioma, lifetime risk ~0.5%) remains a low absolute risk. PGS do not replace family history.

Polygenic scores update (03-10-2026). The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Colorectal cancer: new p92.7 (genomic), p58 and p98 scores; consensus **p80**: confirms high.
- Bladder cancer: new genomics give p57 and p27 and 24 SNPs p86; with the previous (p90, 15 SNPs), the consensus is **p72**. It goes from high to intermediate-high; Slow NAT2, zero tobacco, avoid aromatic amines and pioglitazone are not recommended.
- Prostate (consensus p5.5) and melanoma (p18) confirm low.

Favourable and unfavourable

Favourable

- **No pathogenic variants confirmed** in the hereditary cancer genes covered by the chip and by imputation (BRCA1/2, MLH1/MSH2/MSH6/PMS2, APC, MUTYH, CHEK2, HOXB13, TP53, CDKN2A, PALB2, ATM). The old signals in APC, ATM, NF1 and TSC2 were probe artifacts (impossible homozygotes).
- **Melanoma with low polygenic background:** PGS in the 6.5 (24 variants) and 25 (45 variants); MC1R without any of the 6 variants tested; TYR, IRF4, ASIP and CDKN2A A148T without risk alleles ATM D1853N A/A (2 protective copies).
- **Squamous cell carcinoma at percentile 3.5** and basal cell in 13 (variant 273 score).
- **Kidney (percentile 0.8), non-Hodgkin lymphoma (1.1), Hodgkin (0.8)** and testis (24, KITLG A/A without risk allele).
- **Lung at the 23rd percentile** (with caution: CHRNA5 and TERT weigh mainly in smokers).
- **Prostate without strong alleles:** HOXB13 G84E absent; the 3 loci 8q24 with the highest effect without risk alleles (rs1447295 CC, rs16901979 CC, rs4242382 GG).
- **Colorectal:** APC I1307K, MUTYH Y179C/G396D and CHEK2 I157T absent; EIF3H and GREM1 without risk alleles; the two small scores of this table are medium (49 and 56), but the complete genomic score is high (p94.3; see the polyps report).
- **The probable blood group** (ABO rs505922 T/T): associated with less pancreatic cancer.

Unfavourable

- **Bone at the 90th percentile** + NAT2 slow acetylator (rs1495741 A/A) + MYC 8q24 T/T: more sensitivity to aromatic amines (tobacco, smokes, paints, very burned meat).
- **Glioma at the 91st percentile:** TERT rs10069690 T/T, RTEL1 rs6010620 G/G and TERT rs2853676 heterozygous. Absolute risk remains low.
- **Pancreas at the 83rd percentile** (NR5A2 A/A, KLF5 and CLPTM1L heterozygotes), partially compensated by the probable O group.
- **Esophagus at the 85th percentile** (not very relevant with zero alcohol) and thyroid in 79 (FOXE1 and NKX2-1 heterozygotes).
- **TERT axis:** rs2736100 C/C and rs10069690 T/T (longer telomeres): higher risk of some cancers and, more likely, more prone to clonal hematopoiesis (CHIP). Cellulose activators.
- **Chronic lymphocytic leukemia at the 69th percentile** and basal cell in 70 in the MTAG score (discordant of the main score, the 13th percentile).
- **Diffuse gastric:** PSCA rs2294008 T/T (2 copies, ~1.3× per copy).
- **SERPINA1 Pi*MZ** (Z carrier): lung and liver more sensitive to smoking and pollution.

Contradiction with a previous report: a previous report placed melanoma at the 89th percentile. With the DNA analyzed here, two independent PGS give percentiles 6.5 and 25 and the strong-effect pigmentation genes are without risk alleles. It is more likely that the polygenic fundus of melanoma is low to medium. Skin surveillance is maintained due to the photosensitivity of isotretinoin, not by genetics.

Colorectal

Update: The PGS000074 and PGS000055 scores below (49 and 56 percentiles) are small. The complete genomic score (colorectal polyps report) gives p94.3 and p78.5, with early onset p82.3 and polyps p92, 7. The complete score prevails: the colorectal polygenic risk is high.

Colorectal cancer is the most influenced by modifiable factors (fiber, processed meat, exercise, visceral fat). The SNPs below are the loci of GWAS more replicated; individually they have effects of 1.05–1.25× per copy. The high penetrance genes (APC I1307K, MUTYH) are without variants. Lynch syndrome (MLH1/MSH2/MSH6/PMS2) is not reliably detectable in a chip: no pathogenic ClinVar variant was found in these genes, but only sequencing excludes it.

Interpretation: 9 Risk loci with 1–2 copies (BMP4 C/C, LAMA5 C/C, 8q24, SMAD7, COLCA2, 10p14, 20p12) balanced by EIF3H and GREM1 without risk. PGS is in the middle (49 and 56 percentiles): medium polygenic risk. This confirms that the 76th percentile of a previous report does not reproduce in the raw DNA (§11 profile).

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000074	Colorectal cancer (103 Graff 2021 variants)	48 7 (medium)	-0 07	typical	100 of 103	92.6%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000055	Colorectal cancer (76 variants) already calculated in the §20 profile	55 6 (medium)	0 17	typical	74 of 76	97.1%	high	Known susceptibility variants (genome-wide signifier SNPs); PGP000040, PMID 29917119

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (5)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
EIF3H (8q23)	8q23.3	rs16892766	AA	chip	C	0	favourable	C allele: colorectal risk ~1.25× per copy [strong]
GREM1/SCG5	15q13	rs4779584	CC	chip	T	0	favourable	T allele: plus GREM1 (antagonist BMP), colorectal risk ~1.2× [moderate]
APC	APC I1307K	rs1801155	TT	chip	A	0	favourable	Allele A (I1307K, rare in the reference population): colorectal risk ~1.5–2×; T/T : non carrier [strong]
MUTYH	MUTYH Y179C	rs34612342	TT	chip	C	0	favourable	Allele C (Y179C): carrier of MAP (polyposis associated with MUTYH); biallelic = high risk, monoallelic : mild risk [strong]
MUTYH	MUTYH G396D	rs36053993	CC	chip	T	0	favourable	T (G396D) allele: idem (MAP) [strong]

Unfavourable (7)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
MYC/CCAT2 (8q24)	8q24 rs6983267	rs6983267	GT	chip	G	1	unfavourable	G allele: more binding of TCF4 to MYC enhancer; colorectal and prostate risk ~1.2× per copy [strong]
SMAD7	Intronic SMAD7	rs4939827	TC	chip	T	1	unfavourable	T allele: minus SMAD7 (via TGF-β/BMP), colorectal risk ~1.18× [strong]
	11q23.1	rs3802842	CA	chip	C	1		

COLCA2 (11q23)							unfavourable	C allele: colorectal risk ~1.1× [moderate]
10p14	10p14	rs10795668	GA	chip	G	1	unfavourable	G allele: colorectal risk ~1.1× (A protector) [moderate]
BMP4	BMP4	rs4444235	CC	chip	C	2	unfavourable	C allele: colorectal risk ~1.1× [moderate]
20p12 (BMP2)	20p12.3	rs961253	CA	chip	A	1	unfavourable	Allele A: colorectal risk ~1.1× [moderate]
LAMA5	20q13.33	rs4925386	CC	chip	C	2	unfavourable	C allele: colorectal risk ~1.08× (protective T) [moderate]

Prostate

Two scores with different results: that of 161 variants (Graff 2021, 93% of weighted coverage) gives the 70th percentile; that of 269 variants (Conti 2021, multiethnic, 76% of weighted coverage) gives percentile 1.3. The difference comes from the variants with the greatest effect that is missing in the second. Conservative reading: medium to slightly above average polygenic risk, without strong alleles (HOXB13 G84E, 8q24 2 and 3 regions). MSMB and HNF1B were left out because the imputation has low confidence in these SNPs.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000084	Prostate cancer (161 Graff 2021 variants) Main score (higher coverage)	69 8 (above the average)	0 57	unfavourable	147 of 161	93.5%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000662	Prostate cancer (269 variants Conti 2021) discordant from the previous one; lower weighted coverage	1 3 (low)	-2 16	favourable	221 of 269	75.6%	moderate	Independent genome-wide signifiers identified by fine-mapping using JAM (Joint Analysis of Marginal summary statistics); PGP000122, PMID 33398198

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (5)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
HOXB13	HOXB13 G84E	rs138213197	CC	chip	T	0	favourable	T allele (G84E): risk of prostate cancer ~3-5× <i>[strong]</i>
8q24 (3 region)	8q24.21	rs1447295	CC	chip	A	0	favourable	Allele A: risk ~1.4× <i>[strong]</i>
8q24 (2 region)	8q24.21	rs16901979	CC	chip	A	0	favourable	Allele A: risk ~1.5× <i>[strong]</i>
8q24	8q24.21	rs4242382	GG	chip	A	0	favourable	Allele A: risk ~1.4× <i>[strong]</i>
CHEK2	CHEK2 I157T	rs17879961	AA	chip	G	0	favourable	Allele G (I157T): CHEK2 of partial function; mild risk (prostate, breast, colon) <i>[moderate]</i>

Unfavourable (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
NUDT11 (X)	Xp11.22	rs5945619	C	chip	C	1 of 1 (X)	unfavourable	C allele: risk ~1.2× (chromosome X, 1 copy in man) <i>[moderate]</i>
KLK3	KLK3/ KLK2	rs2735839	AG	chip	G	1	unfavourable	G allele: risk ~ 1.18× (GWAS Catalog) <i>[moderate]</i>

SNP not found in the data (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
MSMB	MSMB promoter	rs10993994	—	SNP not found in the data (low-	T	—	not found	T allele: less β-microseminoprotein, risk ~1.25× <i>[strong]</i>

				confidence imputation: 0.36)				
HNF1B	HNF1B	rs4430796	—	SNP not found in the data (low- confidence imputation: 0.24)	A	—	not found	Allele A: prostate risk ~ 1.2× (and less diabetes type 2) <i>[strong]</i>

Skin: melanoma, basal cell and squamous cell

The risk of skin cancer depends mainly on pigmentation (MC1R, HERC2/OCA2, SLC45A2, TYR, IRF4, ASIP) and sun exposure. None of the strong effect MC1R variants is present. There is a copy of the common light pigmentation alleles in HERC2 (rs12913832 A/G) and SLC45A2 (C/G; the dark C allele is protective). MTAP rs7023329 A/A (plus nevi) and MX2 heterozygote add little. ATM D1853N A/A is protective.

Interpretation: melanoma with low polygenic background (6.5 and 25 percentiles), very low squamous cell (3.5), low to medium basal cell (13 in the main score; 70 in the MTAG score, which mixes traits of pigmentation). Isotretinoin increases photosensitivity and rapamycin is associated with less non-melanoma skin cancer.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000079	Melanoma (24 Graff 2021 variants)	6 5 (low)	-1 43	favourable	22 of 24	94.2%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000743	Melanoma (45 variants)	25 1 (below the average)	-0 64	favourable	44 of 45	97%	high	Genome-wide signifiers; PGP000151, PMID 29890168
PGS003417	Basal cell carcinoma (273 variants)	13 (low)	-1 13	favourable	264 of 273	95.7%	high	Genome-wide signifier SNPs; PGP000429, PMID 36496446
PGS003416	Basal cell carcinoma (462 variants MTAG) includes pigmentation variants and other correlated traits	69 8 (above the average)	0 52	unfavourable	440 of 462	91.8%	high	Genome-wide signifier SNPs; PGP000429, PMID 36496446
PGS005209	Squamous cell carcinoma (20 variants)	3 5 (low)	-1 81	favourable	20 of 20	100%	high	Genome-wide signifier SNPs; PGP000729, PMID 38454124

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (13)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
MC1R	MC1R R151C	rs1805007	CC	chip	T	0	favourable	T allele (R151C, variant "R"): light/red skin, less eumelanin; melanoma ~2× per copy <i>[strong]</i>
MC1R	MC1R R160W	rs1805008	CC	chip	T	0	favourable	Allele T (R160W, "R"): idem <i>[strong]</i>
MC1R	MC1R D294H	rs1805009	GG	chip	C	0	favourable	Allele C (D294H, "R"): idem <i>[strong]</i>
MC1R	MC1R R142H	rs11547464	GG	chip	A	0	favourable	Allele A (R142H, "R") <i>[strong]</i>
MC1R	MC1R D84E	rs1805006	CC	chip	A	0	favourable	Allele A (D84E, "R") <i>[strong]</i>
MC1R	MC1R V92M	rs2228479	GG	chip	A	0	favourable	Allele A (V92M, "r"): weak effect <i>[weak]</i>
TYR	TYR R402Q	rs1126809	GG	imputation DR2=1.00	A	0	favourable	Allele A (R402Q, thermolabile tyrosinase): melanoma ~1.2× <i>[moderate]</i>

TYR	TYR	rs1393350	GG	chip	A	0	favourable	Allele A: melanoma ~1.2× [moderate]
IRF4	IRF4 intron 4	rs12203592	CC	chip	T	0	favourable	T allele: minus IRF4/tyrosinase, freckles and sensitivity to sun [moderate]
ASIP	ASIP/PIGU (20q11)	rs4911414	GG	chip	T	0	favourable	T allele: less eumelanin, sunburn [moderate]
CDKN2A	CDKN2A A148T	rs3731249	CC	chip	T	0	favourable	T allele (A148T): melanoma risk ~1.5× [moderate]
ATM	ATM D1853N	rs1801516	AA	chip	A (protector)	2	favourable	Allele A (Asn1853): melanoma ~0.85× per copy [moderate]
TP53	TP53 3'UTR (poly-A signal)	rs78378222	TT	chip	C	0	favourable	C allele (rare): less TP53; risk of basal cell, glioma and prostate [moderate]

Unfavourable (4)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
HERC2/OCA2	HERC2 (eyes)	rs12913832	AG	chip	G	1	unfavourable	G allele: light eyes/less melanin; melanoma slightly more likely [moderate]
SLC45A2	SLC45A2 L374F	rs16891982	CG	chip	G	1	unfavourable	G (Leu374) allele: clear pigmentation; C (Phe) protector [moderate]
MTAP/CDKN2A	MTAP	rs7023329	AA	chip	A	2	unfavourable	Allele A: melanoma ~1.18× (common, ~50%) [moderate]
MX2	MX2	rs45430	CT	chip	T	1	unfavourable	T allele: melanoma ~1.12× [weak]

Pancreas

the 83rd percentile in the 22 variant score. They contribute NR5A2 (A/A, major allele), KLF5 13q22 and CLPTM1L (5p15, telomerase locus). Against: ABO rs505922 T/T suggests group O (less pancreatic cancer) and BRCA2 K3326X absent. The pancreas is a rare cancer (life-long risk ~1.5%); an the 83rd percentile corresponds to a relative risk of ~1.3–1.5×. The main modifiable factors are tobacco, diabetes and visceral fat.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000083	Pancreatic cancer (22 Graff 2021 variants)	83 1 (high)	0 95	unfavourable	21 of 22	96.8%	high	Genome-wide signifiers; PGP000050, PMID 33579919

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
ABO	ABO (group O)	rs505922	TT	chip	C	0	favourable	C allele: non-O group; T/T suggests group O (less pancreatic cancer) <i>[strong]</i>
BRCA2	BRCA2 K3326X	rs11571833	AA	chip	T	0	favourable	T allele (K3326X, C-terminal truncation): mild risk of lung and pancreas (non-classical BRCA2 mutation) <i>[moderate]</i>

Unfavourable (3)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
CLPTM1L/ TERT	5p15.33	rs401681	CT	chip	T	1	unfavourable	T allele: pancreas ~1.19× (but lung ~0.87×: pleiotropic effect of locus TERT/CLPTM1L) <i>[moderate]</i>
NR5A2	NR5A2	rs3790844	AA	chip	A	2	unfavourable	Allele A (majority, ~77%): pancreas ~1.23× <i>[moderate]</i>
KLF5 (13q22)	13q22.1	rs9543325	CT	chip	C	1	unfavourable	Allele C: pancreas risk ~1.2× <i>[moderate]</i>

Lung and respiratory tract

the 23rd percentile. CHRNA5 D398N (1 copy) increases nicotine dependence and risk especially in smokers. TERT rs2736100 C/C (2 copies) is also an adenocarcinoma factor in non-smokers. SERPINA1 Pi*MZ (Z carrier) makes the lung more sensitive to smoking, vaping and pollution (emphysema).

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000078	Lung cancer (109 Graff 2021 variants)	23 2 (below the average)	-0 75	favourable	98 of 109	89.7%	high	Genome-wide signifiers; PGP000050, PMID 33579919

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Unfavourable (3)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
CHRNA5	CHRNA5 D398N	rs16969968	GA	chip	A	1	unfavourable	Allele A: effect mainly on smokers [strong]
TERT	TERT intron 2	rs2736100	CC	chip	C	2	unfavourable	C allele: longer telomeres and risk of lung adenocarcinoma and glioma ~1.2× [strong]
SERPINA1	SERPINA1 Pi*Z	rs28929474	CT	chip	T	1	unfavourable	T allele (Pi*Z): carrier MZ; emphysema with smoking/pollution; lung cancer slightly more likely [strong]

Other cancers (bex, kidney, testis, thyroid, glioma, esophagus, hematological)

Beach (percentile 90): The highest score of this report. It is added slow NAT2 (A/A, NAT2*5/*5 confirmed in the profile) and MYC 8q24 T/T. The effect of slow NAT2 is manifested mainly with exposure to aromatic amines (tobacco, industrial fumes, old hair paints, very burned meat).

Glioma (the 91st percentile): TERT rs10069690 T/T, RTEL1 G/G, TERT rs2853676 heterozygous; CCDC26 without the strong risk allele (IDH-mutated). Low absolute risk.

Thyroid (79), esophagus (85), CLL (69); kidney (0.8), lymphomas (≈ 1) and testis (24) low; myeloma (58) medium. PSCA T/T: 2 copies of diffuse gastric cancer allele.

Polygenic scores of this section

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000071	Bladder cancer (15 Graff 2021 variants)	90 4 (high)	1 26	unfavourable	14 of 15	94.7%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000076	Kidney cancer (19 Graff 2021 variants)	0 8 (low)	-2 46	favourable	16 of 19	86.1%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000086	Testis tumor (52 Graff 2021 variants)	23 9 (below the average)	-0 70	favourable	48 of 52	90.3%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000087	Thyroid cancer (12 Graff 2021 variants)	79 (above the average)	0 82	unfavourable	12 of 12	100%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS002302	Glioma (28 variants)	90 8 (high)	1 37	unfavourable	28 of 28	100%	high	Genome-wide signifiers; PGP000328, PMID 32588423
PGS002298	Esophageal cancer (14 variants) mainly squamous cell carcinoma very linked to alcohol and tobacco	84 8 (high)	1 03	unfavourable	13 of 14	97.5%	high	Genome-wide signifiers; PGP000328, PMID 32588423
PGS000081	Oral cavity cancer and pharynx (14 variants) Low coverage	42 3 (medium)	-0 08	typical	9 of 14	54.6%	low	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000077	Chronic lymphocytic leukemia (75 Graff 2021 variants)	68 9 (above the average)	0 53	unfavourable	69 of 75	88.7%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000080	Non-Hodgkin lymphoma (19 Graff 2021 variants)	1 1 (low)	-1 75	favourable	18 of 19	95.8%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS003449	Hodgkin's lymphoma (21 variants) moderate coverage	0 8 (low)	-2 66	favourable	16 of 21	67.5%	moderate	Genome-wide signifier SNPs; PGP000448, PMID 36273105

PGS002306	Multiple myeloma (23 variants)	57.5 (medium)	0 20	typical	23 of 23	100%	high	Genome-wide signifiers; PGP000328, PMID 32588423
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* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 $\geq$ 0.8, dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Favourable (2)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
KITLG	KITLG	rs995030	AA	chip	G	0	favourable	G allele: testis ~2.5× [strong]
CCDC26 (8q24)	CCDC26	rs55705857	AA	chip	G	0	favourable	G allele: glioma IDH-mutated ~6× [strong]

Unfavourable (9)

Gene	Variant	rsID	Genotype	Source	Effect allele	Copies	Interpretation	Allele effect
MYC (8q24, bladder)	8q24.21	rs9642880	TT	chip	T	2	unfavourable	T allele: bladder cancer ~1.2× [moderate]
TP63	TP63	rs710521	TC	chip	T	1	unfavourable	T allele (strand +; "A" in the original study strand: bladder ~1.15× [moderate]
NAT2	NAT2 (acetylator)	rs1495741	AA	chip	A	2	unfavourable	Allele A: slow acetylator marker (NAT2*5/*5 confirmed by other SNPs) [strong]
PSCA	PSCA	rs2294008	TT	chip	T	2	unfavourable	T allele: diffuse gastric cancer ~1.3× (and less bladder) [moderate]
FOXE1	FOXE1/PTCSC2	rs965513	AG	chip	A	1	unfavourable	Allele A: thyroid cancer ~1.7× [strong]
NKX2-1/PTCSC3	14q13.3	rs944289	CT	chip	T	1	unfavourable	T allele: thyroid cancer ~1.4× [strong]
RTEL1	RTEL1	rs6010620	GG	chip	G	2	unfavourable	G allele: glioma ~1.3× (telomeric helicase) [moderate]
TERT	TERT intron 4	rs10069690	TT	chip	T	2	unfavourable	T allele: risk of several cancers (mamma, ovary, glioma, CLL) [moderate]
TERT	TERT intron 2	rs2853676	TC	chip	T	1	unfavourable	heterozygous: 1 copy of risk whatever the study chain convention [moderate]

Implications for screening (genetic only)

Cancer	Genetic reading	Implication (suggestion, non-applied)
Colorectal	High complete genomic score (p94.3/p78.5; early onset p82.3); without APC/MUTYH Lynch not excluded by the chip	FIT from 40 and colonoscopy to 45 at most (suggestion, non-applied)
Prostate	Low in the complete genomic score (p0.9); the scores of this report differ (70 and 1); without HOXB13	Shared decision of PSA from the 45–50 years, with intervals guided by the base value
Melanoma / skin	Low; without MC1R; isotretinoin	Annual dermatological examination with dermoscopy (by medication and not by genetics); monthly self-examination; photoprotection
Bladder	the 90th percentile + slow NAT2	No validated population screening; blood attention in urine (see soon); avoid smoking and aromatic amines
Pancreas	the 83rd percentile; group O probable	No screening for polygenic risk alone (only for carriers of mutations or family); control glucose and visceral fat
Lung	the 23rd percentile; CHRNA5, TERT, Pi*MZ	Low dose TC only if there is smoking history; avoid smoking, vaping and pollution
Thyroid / glioma	79 / 91	No imaging; neck palpation in routine examination; valuing new neurological symptoms
Hereditary	No P/LP in the covered genes	Clinical germinal panel (sequencing) if there is early or multiple cancer in the family

Polygenic scores (all)

Score	Trait	Percentage*	z	Interpretation	Variants used	Weighted coverage	Reliability	Method / publication
PGS000074	Colorectal cancer (103 Graff 2021 variants)	48 7 (medium)	-0 07	typical	100 of 103	92.6%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000055	Colorectal cancer (76 variants) already calculated in the §20 profile	55 6 (medium)	0 17	typical	74 of 76	97.1%	high	Known susceptibility variants (genome-wide signifier SNPs); PGP000040, PMID 29917119
PGS000084	Prostate cancer (161 Graff 2021 variants) Main score (higher coverage)	69 8 (above the average)	0 57	unfavourable	147 of 161	93.5%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000662	Prostate cancer (269 variants Conti 2021) discordant from the previous one; lower weighted coverage	1 3 (low)	-2 16	favourable	221 of 269	75.6%	moderate	Independent genome-wide signifiers identified by fine-mapping using JAM (Joint Analysis of Marginal summary statistics); PGP000122, PMID 33398198
PGS000079	Melanoma (24 Graff 2021 variants)	6 5 (low)	-1 43	favourable	22 of 24	94.2%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000743	Melanoma (45 variants)	25 1 (below the average)	-0 64	favourable	44 of 45	97%	high	Genome-wide signifiers; PGP000151, PMID 29890168
PGS003417	Basal cell carcinoma (273 variants)	13 (low)	-1 13	favourable	264 of 273	95.7%	high	Genome-wide signifier SNPs; PGP000429, PMID 36496446
PGS003416	Basal cell carcinoma (462 variants MTAG) includes pigmentation variants and other correlated traits	69 8 (above the average)	0 52	unfavourable	440 of 462	91.8%	high	Genome-wide signifier SNPs; PGP000429, PMID 36496446
PGS005209	Squamous cell carcinoma (20 variants)	3 5 (low)	-1 81	favourable	20 of 20	100%	high	Genome-wide signifier SNPs; PGP000729, PMID 38454124
PGS000083	Pancreatic cancer (22 Graff 2021 variants)	83 1 (high)	0 95	unfavourable	21 of 22	96.8%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000078	Lung cancer (109 Graff 2021 variants)	23 2 (below the average)	-0 75	favourable	98 of 109	89.7%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000071		90 4 (high)	1 26	unfavourable	14 of 15	94.7%	high	Genome-wide signifiers;

	Bladder cancer (15 Graff 2021 variants)							PGP000050, PMID 33579919
PGS000076	Kidney cancer (19 Graff 2021 variants)	0 8 (low)	-2 46	favourable	16 of 19	86.1%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000086	Testis tumor (52 Graff 2021 variants)	23 9 (below the average)	-0 70	favourable	48 of 52	90.3%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000087	Thyroid cancer (12 Graff 2021 variants)	79 (above the average)	0 82	unfavourable	12 of 12	100%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS002302	Glioma (28 variants)	90 8 (high)	1 37	unfavourable	28 of 28	100%	high	Genome-wide signifiers; PGP000328, PMID 32588423
PGS002298	Esophageal cancer (14 variants) mainly squamous cell carcinoma very linked to alcohol and tobacco	84 8 (high)	1 03	unfavourable	13 of 14	97.5%	high	Genome-wide signifiers; PGP000328, PMID 32588423
PGS000081	Oral cavity cancer and pharynx (14 variants) Low coverage	42 3 (medium)	-0 08	typical	9 of 14	54.6%	low	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000077	Chronic lymphocytic leukemia (75 Graff 2021 variants)	68 9 (above the average)	0 53	unfavourable	69 of 75	88.7%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS000080	Non-Hodgkin lymphoma (19 Graff 2021 variants)	1 1 (low)	-1 75	favourable	18 of 19	95.8%	high	Genome-wide signifiers; PGP000050, PMID 33579919
PGS003449	Hodgkin's lymphoma (21 variants) moderate coverage	0 8 (low)	-2 66	favourable	16 of 21	67.5%	moderate	Genome-wide signifier SNPs; PGP000448, PMID 36273105
PGS002306	Multiple myeloma (23 variants)	57 5 (medium)	0 20	typical	23 of 23	100%	high	Genome-wide signifiers; PGP000328, PMID 32588423

* Percentage in the reference population (n=633), calculated with exactly the same variants that were used for Sérgio F (chip + imputation DR2 ≥ 0.8 , dosages). A percentile is not an absolute risk: it indicates the relative position of the polygenic fundus.

Mechanisms of ageing

Attachment of the findings to the mechanisms of objectives for immortality. Only the mechanisms with findings in this theme are listed.

Mechanism (from 84)	Genetic findings	Interpretation
Genomic instability	No P/LP in hereditary repair/cancer genes; ATM D1853N A/A; CHEK2 I157T absent	favourable
DNA Damage and DNA Repair Deficit	MUTYH without variants; Slow NAT2 (more genotoxic aromatic amines); BRCA2 K3326X absent	unfavourable
Hayflick's Telomeres and Limit	TERT rs2736100 C/C and rs10069690 T/T, RTEL1 G/G: longer telomeres with higher risk of glioma/lung/bex	unfavourable
Clonal hematopoiesis (CHIP)	The TERT alleles that extend the telomeres are the main germination factor of CHIP: a slightly higher propensity is more likely	unfavourable
Somatic Mutation Accumulation	Low polygenic background for melanoma, squamous cell, kidney and lymphomas; high for bladder and glioma	typical
Cell Senescence	CDKN2A A148T absent; MTAP/9p21 A/A (plus nevi)	typical
Immunosenescence	PGS of very low lymphomas; CLL at the 69th percentile; MX2 heterozygous	favourable
Chronic Inflammation	PSCA T/T (gastritis and diffuse gastric cancer); SERPINA1 Pi*MZ	unfavourable
Intestinal Barrier dysfunction	PSCA T/T (gastric mucus); high colorectal (p94.3)	typical
Redox imbalance	Pigmentation: 1 copy of light alleles (HERC2, SLC45A2); Functional MC1R (eumelanin protects from oxidative stress UV)	favourable

Practical part: suggestions

All the lines below are suggestion (not applied). They are consistent with the v2 genetic plan and with the current protocol (pause curcumin, luteolin, apigenin and quercetin while ALT is high, as an unapplied suggestion; senolytic pulse only fisetin + quercetin and only with normal ALT; caution with telomerase activators; D3 10 000 IU; omega-3 3.2–4 g/day; ferritin below 100). To discuss with the doctor.

- **suggestion (not applied):** Cautel with telomerase activators (epitalon, TA-65, astagaloside): the TERT alleles of Sérgio F already favor long telomeres and add risk of glioma, bladder and CHIP; pause the epitalon is a suggestion of the plan v2 (not applied; the real stack has not changed).
- **suggestion (not applied):** Rapamycin 4 mg of 6 in 6 weeks: maintain; mTOR is a central pathway in several cancers with high score (bladder, glioma) and rapamycin is associated with less non-melanoma skin cancer.
- **suggestion (not applied):** Slow berth + NAT2: not smoking or vaporizing; avoid combustion smoke and solvents; avoid much burned or smoked meat (heterocyclic amines); abundant water; consult soon if blood appears in the urine.
- **suggestion (not applied):** Fiber: the more, the better (≥ 50 g/day; psyllium and PHGG) and daily crucifers: protect from colorectal and help detoxification of amines (slow NAT2, probable null GSTM1).
- **suggestion (not applied):** Pancreas: keeping visceral fat and glucose low (retatrutide and dapagliflozin already act here); zero alcohol already fulfilled.
- **suggestion (not applied):** Skin: daily photoprotection SPF 50 with isotretinoin; annual skin examination (by medication).
- **suggestion (not applied):** Screening: colorectal from 45 years and PSA by shared decision from 45–50, as in the general population (genetics does not justify anticipating); clinical germinal panel only with family history.
- **suggestion (not applied):** Vitamin D3 10 000 IU and omega-3 3.2–4 g/day: maintaining according to the current protocol; vitamin D is associated with less cancer mortality.
- **suggestion (not applied):** Ferritin below 100: maintain (high iron is pro-oxidant and associated with colorectal and hepatic cancer); without supplementary iron.
- **suggestion (not applied):** LDL and ApoB as low as possible and maximum fiber: no conflict with any oncological finding.

Limitations of array and analysis

- Lynch syndrome, BRCA1/2 and other hereditary syndromes often result from rare or private variants, indels and deletions that the chip does not see. This report does not exclude them.
- The scores of 1-2 dozens of variants (pancreas, bladder, kidney, thyroid) capture little heritability; the percentile is unstable and can change with larger scores.
- Two pairs of scores disagree (prostate: 70 vs 1.3; basal cell: 13 vs 70). In these cases the reading is "medium" until there is a high coverage genomic score.
- MSMB rs10993994 and HNF1B rs4430796 were not genotyped (Imputation DR2 < 0.8).
- The consumption chips genotype ~0.7 M positions; rare variants, indels, repeat expansions and CNVs are not reliable or do not exist in the data. An absence of a rare variant here does not exclude a pathogenic variant: only clinical sequencing (panel, exome or genome) does so.
- Imputation (Beagle 5.5, public panel of complete genomes to 30x) was only used with DR2 ≥ 0.8; below that SNP is treated as "SNP not found in the data".
- Polygenic scores indicate relative position in the reference population, explain a small fraction of risk and do not include lifestyle, medication or family history.

Methods, chain and sources

Data: chip genotypes (GRCh38, strand +) and local imputation with $DR2 \geq 0.8$; all analyzed in the local computer. No genotype was sent out; public databases (Ensembl/dbSNP, GWAS Catalog, PGS Catalog) only consulted public identifiers (rsIDs and IDs of scores).

Strand: All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP. For each SNP the genotype was compared with the alleles of dbSNP: 57 concordant without conversion; 0 needed complement; 2 palindromic SNPs (A/T or C/G), solved by the strand + of the chip and by the allelic frequency. The risk alleles of the studies that were in the opposite strand were converted to the strand + (indicated in the SNP note).

Effect allele: defined by the literature and verified in GWAS Catalog (most significant association for the trait). "Favourable" : without the risk allele or with the protective allele; "unfavourable" : with 1–2 copies of the risk allele; "typical" : without a rare protective allele (as most people).

Polygenic scores: Harmonized coefficients of PGS Catalog (GRCh38). Score = Δ coefficient $\times$ dose of effect allele. Empirical percentile in the reference population ($n=633$) with the same variants. Reliability: high $\geq 85\%$ of weighted coverage, moderate 60–85%, low $<60\%$.

Count: 57 Curated SNPs: 54 chip, 1 imputed, 2 without chip genotype/imputation (0 with previous report genotype).

Main sources

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- Rothman N et al. / García-Closas M et al. NAT2 slow acetylation and bladder cancer. Lancet 2005; Hum Mol Genet 2011.
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- PGS Catalog; GWAS Catalog; dbSNP/Ensembl; ClinVar; public panel of genomes complete to 30x.